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Assessing the Relevance of Groundwater Level as a Predictor Variable in 30-day Stream Discharge Forecasts for the Susquehanna River Basin

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ABSTRACT

Stream discharge is the volume of water flowing through a channel over a specified time period. Stream discharge has major implications for ecological health and human economy, including hydropower generation and agriculture. Forecasting stream discharge enables educated policy decisions as well as flood and drought warnings. Many variations of process-based and data-driven methods have been developed for hydrologic modeling. We investigate two data-driven methods: Long-Short Term Memory Recurrent Neural Networks (LSTM) and AutoRegressive Integrated Moving Average (ARIMA) models. We identify a lack of groundwater inclusion in model development and use these modeling methods to investigate the importance of including groundwater measurements for forecasting in the Susquehanna River Basin (SRB). The SRB spans parts of New York, Pennsylvania, and Maryland to provide 50% of Chesapeake Bay's freshwater input. Discharge forecasting is especially important in the SRB because of the watershed's flood-prone status and importance to the Chesapeake Bay. Models were trained on 14 years of USGS groundwater and stream data as well as NOAA weather station data. Limited hyperparameter tuning was done before the models were fit to the training data, cross-validated on twelve 30-day validation splits, and tested against a final 30-day split. Assessed metrics include Root Mean Squared Error, Mean Absolute Percent Error, Kling-Gupta Efficiency, and Nash-Sutcliffe Efficiency. The LSTM model without groundwater performed the best across all metrics. The ARIMA model without groundwater had superior metrics than the ARIMA with groundwater. These results indicate that including historical average groundwater as a projected input variable does not improve forecasting accuracy for LSTM nor ARIMA models.

INTRODUCTION

I. Relevance of Stream Discharge Forecasting

Understanding and forecasting stream discharge has major implications for human safety, economic concerns, and ecological health. Forecasting refers to predicting future events based on past and present data (Middle Atlantic, 2010; Rajib et al., 2025). Short-term stream forecasting is most essential for flood-risk management. Flooding causes between $179.8 and $496.0 billion in damages annually across the United States (Joint Economic Committee, 2024). Major flooding-related costs include preventative infrastructure upgrades, direct and indirect commercial impact, destruction of commercial physical assets, and damage to homes (Joint Economic Committee, 2024). Many flood damage estimates often neglect economic loss that is difficult to quantify (Allaire, 2018). While flooding interrupts public services and stalls economic production, low-income households suffer disproportionately. Low-income households often have to divert money away from education and health care to spend on flood recovery and prevention (Allaire, 2018). Forecasting floods can help reduce these costs through preventative actions. Preventative actions include evacuation, transporting portable belongings, stockpiling food and water, and scheduling crop harvesting or planting around forecasted floods. Longer timeframes, such as a 10-day forecast, provide a much greater opportunity for preparedness than shorter, more common 3-day forecasts (Webster & Jian, 2011).

The benefits of stream forecasting expand beyond reduced flooding damages. Sophisticated stream discharge forecasts support all water related activities, including hydropower generation, irrigation, and transportation (Girihagama et al., 2022; Fleming & Gupta, 2020). Managing these activities relies on low flow standards, which are set with guidance by the Environmental Protection Agency (EPA, 2025). Low flow is typically defined as

the lowest 7-day average discharge that occurs on average once every 10 years (EPA, 2025). Low flow can create hazardous conditions for aquatic life because less water is available to dilute contaminants (EPA, 2025). Doyle et al. (2005) explains that discharge is related to many ecological processes, directly affecting organic matter transport, algal growth, nutrient retention, macroinvertebrate disturbance, and habitat availability. Discharge can be considered a master variable when analyzing and modeling stream ecosystems (Doyle et al., 2005). Confidently forecasting stream discharge enables well-informed water resource management decisions, as policy makers can increase or decrease water extraction based on projected values.

II. Stream Discharge Hydrology

Stream discharge is influenced by many hydrogeologic factors, which may affect forecasting. In general, stream discharge is based on precipitation and groundwater recharge inputs (Gao et al, 2025). The relationship between stream discharge, precipitation, and groundwater is based on catchment properties and geomorphology of subsurface pathways (Carlier et al., 2018). During times of low precipitation, a stream is primarily fed by groundwater, called baseflow (USGS, 2013). As precipitation falls, it runs off the ground surface into the stream, causing the stream's discharge to rise above its base level in a phenomenon called direct runoff. (USGS, 2013). Direct runoff can follow two general pathways: surface runoff or subsurface runoff. Surface runoff does not infiltrate the ground but travels along the ground surface to the stream. Streams that experience a high proportion of surface runoff are flashier, meaning water level changes quickly. Subsurface runoff refers to precipitation that enters the soil and moves towards the stream via subsurface pathways (Carlier et al, 2018). Subsurface geology governs infiltration rates and therefore determines the proportion of surface

runoff compared to subsurface runoff (Carlier et al., 2018). The dynamics of runoff pathways are influenced by vegetation, soil, topography, and geology. These fundamental characteristics of a catchment basin affect infiltration, evapotranspiration, interception, and overland and subsurface flow (Carlier et al., 2018). Many processes influence how much water input the stream receives as well as when the input reaches the stream. Accounting for this multitude of factors in stream discharge modeling has traditionally required extensive mathematical equations.

III. Classical Stream Discharge Modeling

Historically, stream discharge modeling has been dominated by process-based approaches. Girihagama et al. (2022) categorizes process-based models as conceptual lumped models and physical-process-based distributed models. Conceptual lumped models simplify the physical properties involved in runoff generation through mathematical formulas. The simplifications make conceptual lumped models easier to use and are based on sound principles, but these models do not capture spatial variability of hydrologic processes.

Physical-process-based distributed models capture the spatial variability of the watershed through tailored equations. This watershed-specific tailoring causes physical-process-based distributed models to not be transferable between watersheds (Girihagama et al., 2022). The process-based modeling approach has been applied successfully and is theoretically intact. Process-based modeling has distinct benefits, including high interpretability and prediction of parameters that lack historical data records (Espinoza et al, 2025; Feng et al, 2022). However, process-based modeling struggles with constraints such as oversimplified parameterization, extensive data requirements, and high computational complexity (Hamzeh et al., 2025). With

recent expansions in data collection, data-driven modeling approaches stand as a valuable alternative to process-based modeling.

IV. Data-driven Discharge Modeling

Data-driven models refer to algorithms that learn patterns from existing data. Their main advantage over process-based modeling is that data-driven models do not require detailed knowledge of a stream system's hydrogeologic properties (Geer, 2021). Also, data-driven models can run with less input data, which has led to an increase in their popularity (Hamzeh et al., 2025). Data-driven models include both traditional statistical models as well as deep network machine learning (ML) techniques (Girihagama et al., 2022; Hamzeh et al., 2025; Xu & Liang, 2021). Statistical approaches quantify the relationship between input and output variables, whereas ML techniques process input variables into an artificial neural network to produce a target output.

The most powerful statistical models for hydrology are those that incorporate time-series analysis, such as Autoregressive Integrated Moving Average (ARIMA) models (Girihagama et al., 2022; Hamzeh et al, 2025). ARIMA models rely on long records of historical data to derive relationships among data sequences and forecast future values of a target variable (Girihagama et al, 2022). A basic ARIMA model includes recent target variable values to predict a future target variable value An advanced version of ARIMA is SARIMAX, which allows for the inclusion of seasonal trends (S) and the input of external independent variables (X). For a monthly data set, allowing the seasonal component lets the SARIMAX model look back at the target variable value 12 months ago, therefore enabling it to maintain seasonal consistency. In hydrology, external independent variables include precipitation and temperature (Perazzollo et al., 2025).

The SARIMAX model forecasts future values based on past values of the target variable as well as coefficient values of exogenous input data. The parameter weights allow us to draw some conclusions about a system's hydrogeologic properties. For example, if a SARIMAX model weighs precipitation very strongly relative to groundwater, that would indicate the stream relies on meteorological input more so than groundwater baseflow.

Alternatively, deep network ML models process data and extract non-linear relationships without explicitly defining physical properties. The ability of ML models to learn patterns without defining physical properties improves the model's generalizability to other watersheds, a feature that process-based models and statistical models struggle with (Girihagama et al., 2022). The basic form of deep network ML models are Artificial Neural Networks (ANNs). ANN architecture includes an input layer, one or more hidden layers, and an output layer (Figure 1). Information gets passed from the input layer to the hidden layers, and each neuron in the first hidden layer computes a single output based on the inputs. The neuron outputs then get passed to each neuron in proceeding hidden layers until the output layer produces a comprehensible prediction value. ANNs train on data and adjust neuron weights by starting with the output layer and moving backwards through the hidden layers (Xu & Liang, 2021). While ANNs are powerful tools to process and learn from complex data, the original ANN architecture struggles with time-dependent hydrology data.

The Long-Short-Term-Memory (LSTM) variant of the Recurrent Neural Network (RNN) architecture has been more successful in hydrologic modeling (Cheng et al, 2020; Girihagama et al, 2022; Hamzeh et al, 2025; Xu & Liang, 2021). RNNs have the capability to process variable input and output lengths while keeping data in sequential order. These features are a benefit of RNNs over ANNs for time-dependent hydrologic forecasting, but traditional RNNs have

difficulty retaining information from much earlier sequences (Girihagama et al., 2022). LSTM is a modification to the RNN model that allows for capturing long-range relationships. LSTMs have been used in hydrologic applications for rainfall-runoff modeling, stream discharge predictions for ungauged basins, modelling flash flood events and flood forecasting, and simulating reservoir outflows (Girihagama et al., 2022).

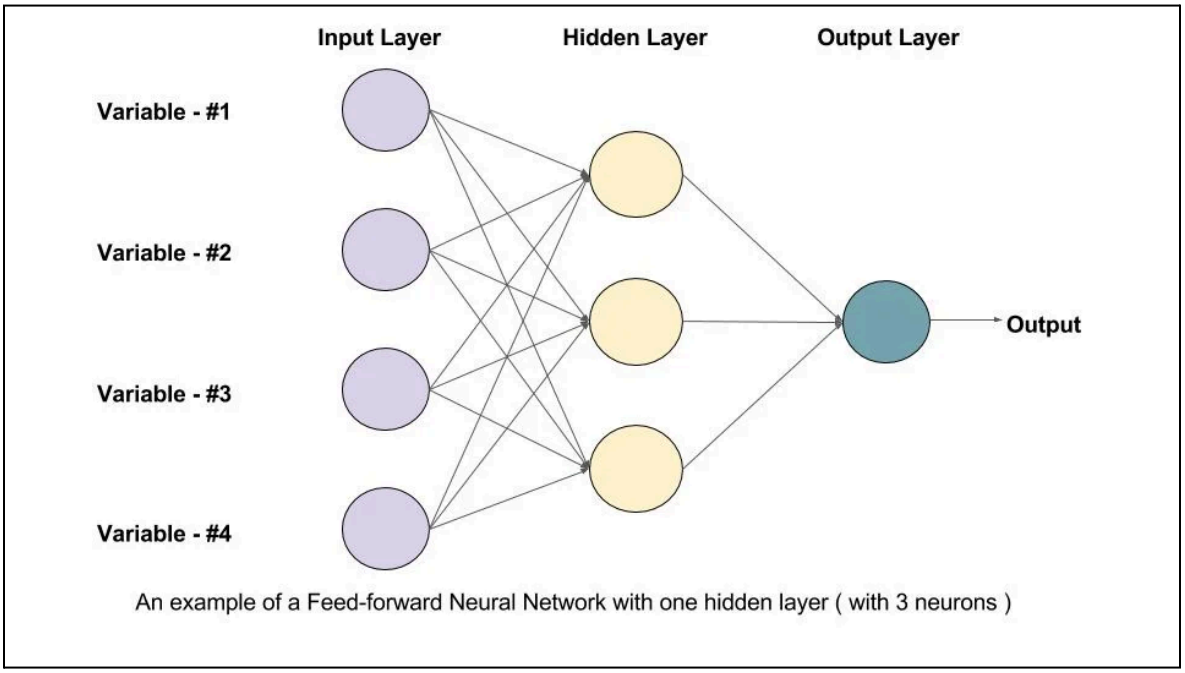


Figure 1: Example Artificial Neural Network Architecture (Gupta, 2017). The example ANN has four input variables in the input layer, four nodes in the hidden layer, and produces a single output. ANNs can have any number of hidden layers and any number of nodes in each hidden layer.

V. Application to Susquehanna River Basin

Many researchers develop models without accounting for groundwater. Galavi et al (2023) develop an ARIMA model which does not use any exogenous input variables and find it to be ineffective at forecasting water level for long time horizons. Hamzeh et al (2025) compare different types of forecasting models using past stream guage and upstream guage data. They

acknowledge that discharge forecasts have long-term dependencies based on precipitation runoff; this long-term impact of precipitation may be captured statistically by groundwater level. Gao et al. (2025) studied the Yanglou catchment in eastern China and found that stream discharge is associated with periods of rapid groundwater rise. High soil permeability in the Yanglou catchment suggests that overland flow is not the primary runoff transport pathway. The research in the Yanglou catchment suggests that a stream's response to groundwater levels is based on the underlying geology of the catchment.

The relationship between groundwater and stream discharge forecasting can be further explored in the Susquehanna River Basin (SRB). The SRB is an ecologically and economically valuable watershed that spans through upstate New York, most of eastern Pennsylvania, and a small section of Maryland. The SRB provides 50% of the freshwater input to the Chesapeake Bay, which is the United States' largest and most economically valuable estuary (Leathers et al, 2018). The SRB's headwaters in New York and Pennsylvania were carved by glacial movements and deposits. Glacial action created many bedrock and valley-fill aquifers, which are valuable human water resources (Heisig & Fleisher, 2022). The SRB is a particularly effective area for investigating the role of groundwater in forecasting models because sand-and-gravel aquifers provide groundwater discharge during low flow periods (Randall, 2010) Many sites in the northern SRB are incised in impermeable till or bedrock, which prevent groundwater from flowing anywhere except into the stream (Randall, 2010). This suggests that stream discharge forecasts of the northern SRB would highly rely on groundwater levels. While the New York section of the SRB has been researched in detail, the Pennsylvania and Maryland portions lack descriptions of groundwater hydrology.

VI. Hypothesis

By building forecasting models for the SRB, we can gain insight into its localized hydrogeologic characteristics. For each study site, we will build four models. We will train two SARIMAX and two LSTM models to forecast future stream discharge values based on hydrologic and meteorologic inputs. We aim to study the impact of groundwater data on discharge forecasting. As such, one SARIMAX and one LSTM model for each site will not include groundwater level measurements to act as controls to compare with the groundwater-included models. We will develop both SARIMAX and LSTM models because SARIMAX has more interpretable parameter weights, while LSTM models are more commonly used in other hydrology modeling research. We hypothesize that both SARIMAX and LSTM models that include groundwater as an input will forecast stream discharge of SRB tributaries more accurately than models that do not include groundwater. Forecasting accuracy will be measured by root mean squared error (RMSE), Mean Absolute Percent Error (MAPE), the Kling-Gupta Efficiency (KGE), and the Nash-Sutcliffe Efficiency (NSE).

METHODS

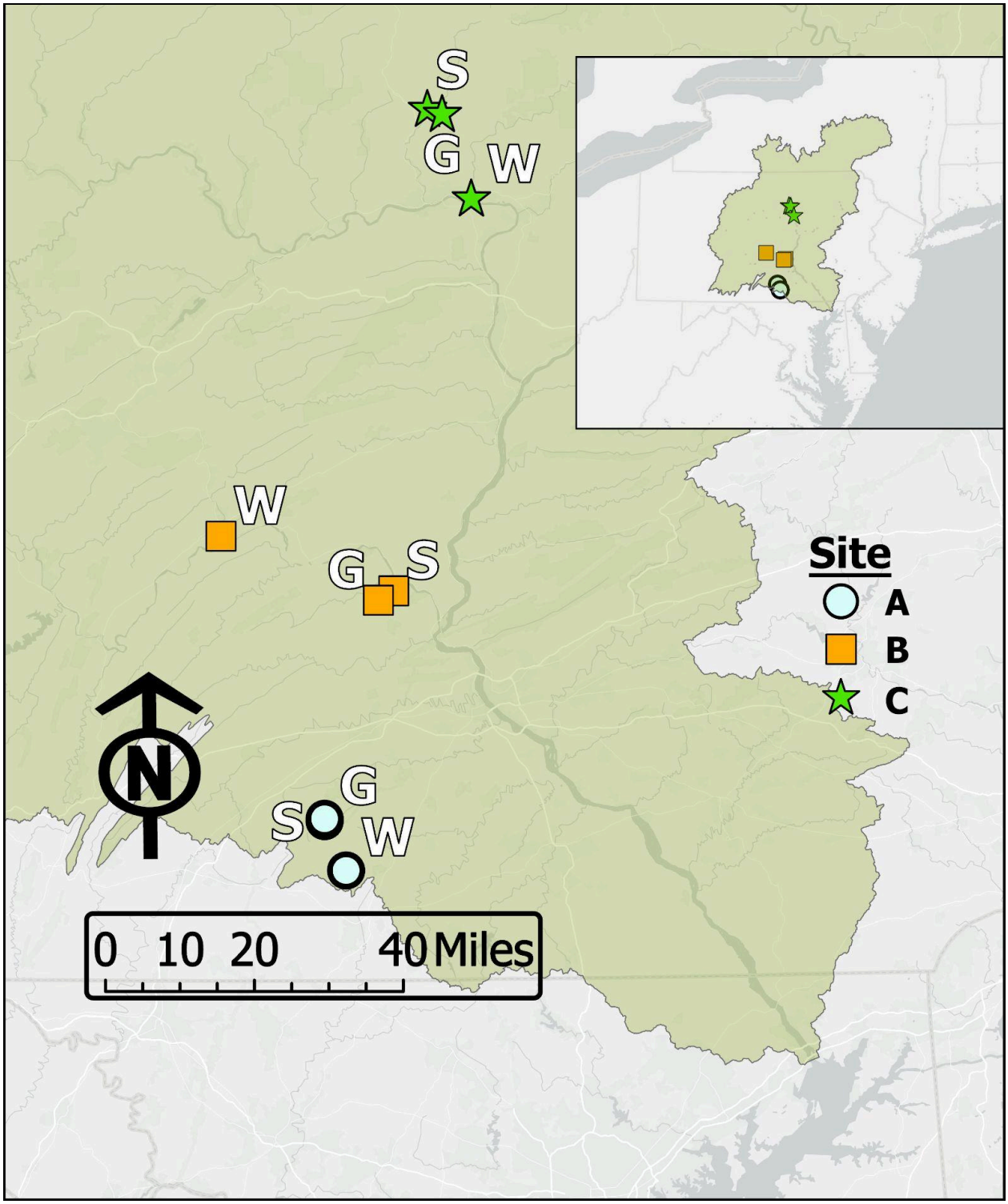
I. Susquehanna River Basin, Site Selection

Selecting sites in the SRB depended on monitoring instrument placement and data availability gathered from the United States Geological Survey (USGS) and the National Oceanic and Atmospheric Administration (NOAA). That is, three instruments needed to be in close proximity: a USGS stream gauge, a USGS groundwater well, and a NOAA weather station (USGS, 2026; NOAA, 2026). Additionally, we preferred sites with extensive data records so as to provide the models with more training information.

Site A is located in and around Biglerville, Pennsylvania in the Lower Susquehanna region (Map 1, Table 2). This geographic region is dominated by metavolcanic rock, which is unique in the SRB (Susquehanna River Basin Commission [SRBC], 2006). Land use is dominated by row cropping with deciduous forest (SRBC, 2021).

Site B is located in and around Lewistown, Pennsylvania in the Juniata region (Map 1, Table 2). This geologic region has a high presence of carbonate rock, including limestone and dolomite, as well as shale. The Appalachian Mountains cut through this region and cause geologic folding (SRBC, 2006). Land use includes high and low levels of urban development, row cropping, and some deciduous forest (SRBC, 2021).

Site C is located around Williamsport, PA in the West Branch region (Map 1, Table 2). This geologic region was heavily impacted by glacial deposit and consists mostly of sandstone and shale (SRBC, 2006). Land use includes high and low levels of urban development as well as row cropping and pasture lands (SRBC, 2021),



Map 1: Map of study sites and monitors within sites. (S = stream gauge; G = groundwater monitor; W = weather station). Monitor ID codes and exact coordinate locations are provided in Table XX.

II. Code Development

All code functions were developed in Python 3.12 using Jupyter Notebooks, GitHub, and GitHub Codespaces. We used Jupyter Notebooks to experiment with model design as well as familiarize myself with the USGS and NOAA data portals. Then, we converted the code into GitHub and GitHub Codespaces for final edits, running, and shareability. Required external packages included pandas, numpy, plotly, matplotlib, seaborn, sktime, scikit-learn, dataretrieval, pmdarima, and permetrics (The pandas…, 2026; Harris et al, 2020; Plotly Technologies, 2015; Hunter, 2007; Webster & Jian, 2021; Király et al, 2025; Pedregosa et al, 2011; Hodson et al, 2023; Smith, 2017; Van Thieu, 2024)

USGS data included two key parameters: stream discharge and groundwater depth below ground surface. These were accessed using the dataretreival package managed by USGS (Hodson et al, 2023). The stream discharge data used data code 00060, or the daily mean discharge in cubic feet per second. The groundwater level used code 72019, which often returned three values for each day: maximum, minimum, and mean feet below ground level. We selected the mean groundwater level and created a data frame with date, groundwater level, and stream discharge columns.

Because statistical models train more accurately with normalized data, we included a function that checked the skew value of groundwater level and stream discharge (Geer 2021). If the skew for either variable was greater than 0.75, the variable underwent logarithmic transformation. In the case of a negative skew and skew less than -0.75, which was often the case with groundwater level, the values were reflected before logarithmic transformation. For groundwater, this changed the meaning of the variable from groundwater level in feet below ground surface to groundwater level in feet above historic groundwater minimum.

For weather data access,weselected the closest weather station to the stream and groundwater monitors that had sufficient data records. We downloaded data for each weather station and read it into the program, selecting for the following daily variables: maximum temperature, precipitation, snow, and snow depth. The program selected weather data for the given start and end dates. Then, the program merged weather data and hydrology data into one dataframe. Because the statistical models cannot run with missing data values, we implemented an interpolation step. Missing values were first interpolated linearly. That is, if one data point was missing between two known data points, then missing data point became the mean of its two neighbors. Also, the data were backfilled and forward-filled. If data were missing at the start or end of the dataframe, the missing values were set equal to the first or last actual data point, respectively.

Following the dataframe setup and cleaning, the data was prepared for the models. We created a copy of the dataframe excluding the groundwater level column, thus establishing a groundwater-included dataframe and a groundwater-absent dataframe. Each dataframe underwent identical processes: splitting into training, validation, and test sets. Then, we fit each split dataframe with an LSTM model and with a SARIMAX model. The test data was split from the rest of the data by separating the last 30 days in the dataset. The test sets were not seen by the model during training, but were left for calculating final forecast performance metrics. For actual model tuning, the training/validation data was split using an expanding window splitter method. The windows were each 30 days in length, and there were 12 splits. Therefore, the first training fold spanned from the established start data (January 1st, 2010) to 360 days before the final point in the training/validation set. Each model trained on this first training set, T_1, established parameter values, and scored its parameter values against the first validation set, V_1, or the next

30 days of data. Then, the model moved onto the next training and validation sets, T_2 and V_2. T_2 can be thought of as $T_1 + V_1$. So, the second training set includes all of the points in the first training set and all of the points in the first validation set. V_2 is the next set of 30 days after the end of T_2. This pattern repeats itself so that the model trains against all seasons in a year.

III. LSTM

The LSTM was set up using the *sktime* NeuralForecastLSTM() function (Sktime, 2025). The function required specified hyperparameters: a minmax scaler, an input size of 365, a batch size of 32, a learning rate of 0.005, and 200 maximum steps, 2 hidden layers, and 128 units per layer. The minmax scaler sets all input variables on a range between 0 and 1, where 0 is the minimum value of that variable, and 1 is the maximum value; all other values are set proportionally between the minimum and maximum. The minmax scaler is more sensitive to outliers which is important for capturing occasional high discharge events (Amorim at al., 2022). The input size of 365 lets the LSTM look back at the past year of data when forecasting a future point. Limiting the LSTM's exposure to past data helps prevent long model training times and enables the LSTM to learn new data patterns instead of relying too strongly on annual cycles. Allowing for 365 days of input still allows the model to pick up some annual signal. Batch size is the number of data points the model looks at before adjusting its parameter values. A batch size of 32 is relatively small, which couples well with the low learning rate of 0.005. The small batch size means that the LSTM adjusts its parameters based on a small, potentially very noisy subset of data, but the low learning rate prevents the parameter values from adjusting too drastically. This combination helps capture the variability in hydrology data. The 200 maximum steps means the LSTM passes over 200 batches of data, which allows the model the opportunity to adjust its

parameter values while keeping computational runtime low. Finally, the 2 hidden layers of 128 units help the model capture complex relationships without being overly complex, so as to prevent overfitting.

The LSTM trained its parameters against the 12 cross-validation splits. Each fit was scored using root mean squared error (RMSE), and the model selected the best model weights.The selected LSTM forecasted the final 30 days of data using estimates for future predictor variables. These estimates were calculated as historical averages for each parameter value by calendar date, and this forecast was scored against the 30-day test set.

IV. SARIMAX

To set up the SARIMAX model, we relied on the sktime AutoARIMA function, which is built on pmdarima. These ARIMA models required the hyperparameters: p, d, and q as well as seasonal versions P, D, and Q. Other hyperparameters were period of seasonal differencing, model evaluation information criterion, and inclusion of stepwise fitting algorithm. Lowercase p is the autoregressive term, which adds a weight based on a recent data point. The maximum p allowed was 2, meaning the model could use up to the past 2 data points as a basis for the next forecast. To avoid high computational costs, maximum seasonal P was set to 0. Lowercase d allows for differencing between points to keep the data stationary. Maximum lowercase d was set to 1 and maximum seasonal D was set to 1. These differencing values allow the model to remove trend and seasonality to predict values more consistently. Lowercase q is the moving average term, which allows the model to use the error in the most recent prediction to forecast the next prediction value. Maximum lowercase q was set to 2, and maximum seasonal Q was set to 0 to avoid high computational costs. The period of seasonal differencing was set to 365, to indicate

365 days pass before a seasonal pattern repeats. The model selector was set to AICc, or Akaike Information Criterion corrected for small sample size. The stepwise fitting algorithm was set to true, which speeds up the model fitting process and is less likely to overfit than other fitting algorithms.

Cross-validation for SARIMAX was purely informational. The model ran for each of the cross validation folds, and for each fit, produced an RMSE score. This allows us to view how the SARIMAX performs across different seasons as opposed to solely the test set. Then, the SARIMAX was refitted on all of the training and validation data. The SARIMAX model predicted the test set for 30 days by relying on daily historical average values as estimated for predictor variables.

V. Model Scoring

Each site had four sets of scoring metrics. The scoring metrics calculated included Root Mean Squared Error (RMSE), Mean Absolute Percent Error (MAPE), Nash-Sutcliffe Efficiency (NSE), and Kash-Gupta Efficiency (KGE). For each site, these metrics were calculated for the groundwater-included LSTM, groundwater-included ARIMAX, groundwater-absent LSTM, and groundwater-absent ARIMAX. The scoring metrics provide an estimate on overall performance of a site's forecast per model. To assess the ability of each model setup to forecast within the SRB generally, the performance metrics were averaged across all sites.

The ARIMAX model is especially important when assessing the contribution of groundwater to model performance because ARIMAX models include parameter weights for predictor variables. It will be noted if the ARIMAX model adds significant weight to the groundwater level as a predictor variable.

An important aspect of hydrologic forecasts is accuracy as time into the future increases, or avoidance of forecast degradation. To measure this, we calculated the point-wise absolute difference between forecasted values and actual values. Therefore, each model for each site has an array of 30 absolute difference values. These were also averaged across sites, to produce a point-wise mean absolute difference per model type. A rate of change and R^2 value were calculated for each model's mean absolute difference set. A positive rate of change would indicate forecast reliability decreases further into the future.

RESULTS

Mean performance metrics were calculated for each of the four model variations (Table 1). The groundwater-absent LSTM performed the best across all metrics; LSTM absent had the lowest RMSE and MAPE as well as the least negative KGE and NSE. The ARIMA absent model performed better than the ARIMA groundwater model in all metrics except KGE. The LSTM groundwater model consistently performed the worst.

Table 1: Mean performance metrics for each model type, with each value representing the mean across all SRB sites (n=3). The groundwater-absent LSTM has the lowest RMSE and MAPE and the least negative KGE and NSE. RMSE = Root Mean Squared Error; MAPE = Mean Absolute Percent Error; KGE = Kling-Gupta Efficiency; NSE = Nash-Sutcliffe Efficiency.

Model Type	**RMSE**	**MAPE**	**KGE**	**NSE**
LSTM GW	0.94	0.32	-0.66	-5.21
LSTM absent	0.52	0.10	-0.08	-0.13
ARIMA GW	0.80	0.20	-0.21	-2.28
ARIMA absent	0.69	0.13	-0.35	-0.93

The forecast degradation determination produced a set of mean absolute differences between the forecasted and actual stream discharge values. A set of this type existed for each

model. For LSTM groundwater, the rate-of-change was -0.0013 ($R^2 = 0.005$) (Figure 5). For LSTM absent, the rate-of-change was -0.0141 ($R^2 = 0.552$) (Figure 6). For ARIMA groundwater, the rate-of-change was -0.0092 ($R^2 = 0.057$) (Figure 7). For ARIMA absent the rate-of-change was -0.0032 ($R^2 = 0.014$) (Figure 4) (Figure 8).

All ARIMA models had a non-zero and statistically significant parameter weight for groundwater level (each p-value < 0.001). The groundwater coefficients were -0.1745, -0.4918, and -0.5104 for sites A, B, and C, respectively

DISCUSSION

I. Relevance of Groundwater

The presence or absence of groundwater data in these models corresponds to an observable difference in model performance. The non-groundwater LSTM and ARIMA models have lower RMSE and MAPE values than their groundwater-included counterparts. In other words, the non-groundwater models predict stream discharge values closer to the real values compared to the groundwater models. The decision to use historical daily average groundwater values likely influenced these forecasts. For sites A and B, historic groundwater influenced the model to forecast much higher values, which led to generally poorer fits. The same effect happened with site C, but C experienced a rapid and consistent increase in real discharge (figure 5). Site C's ARIMA without groundwater model forecasted a more typical, low precipitation month and underestimated discharge, thereby leading to a low KGE value. This example calls attention to the important interplay between precipitation and groundwater. For all sites, groundwater level was high during the test month but few precipitation events occurred with

sites A and B. At site C, when real precipitation and real groundwater level more closely align with historic averages, the groundwater-included ARIMA model performs relatively well.

Aside from overall model scores, the ARIMA model's groundwater coefficients suggest that groundwater is very important to actual stream discharge. All of the groundwater coefficients were on the same order of magnitude as the ARIMA terms, so fluctuations in groundwater level had a meaningful impact on predicted values Additionally, each ARIMA model calculated the groundwater coefficient as having a p-value<0.001. The poor model performance, therefore, may be because the 30-day test period (December 3, 2024 – January 1, 2025) had extreme qualities. In further iterations of this question, it would be beneficial to set the models' test split as another month of the year. It is possible that December 2024 had unusual weather for the entire SRB region, and the performance metrics, therefore, may not represent the models' performances for forecasting other seasons of the year.

II. Model Applicability

An important feature of the trained models is their ability to maintain a consistent forecast across the forecast horizon. Many forecasting models operate on a horizon of 1 to 10 days (Galavi et al, 2013; Girihagama et al, 2022; MARFC, 2010; Webster & Jian, 2011). Models optimized for small horizon forecasting often produce more accurate predictions than our models. Developing a small horizon flood forecaster has value for emergency flood preparedness, but pre-flood measures are limited by the amount of advance warning time. Webster & Jian (2011) acknowledge that rural, developing communities benefit from a forecast horizon of ten days. Farmers can use the ten days to relocate livestock and to reorganize their planting and harvesting schedule to avoid ruined crops. The LSTM and ARIMA models here

were trained and validated to produce a low RMSE across the entire thirty-day horizon. While the models are not particularly effective at forecasting in the short term and generally fail to capture extreme events, their absolute error stays consistent throughout the thirty days.

III. Comparisons to Other Model

Comparing models across research is made difficult because RMSE, KGE, and NSE are dependent on the scale of the data. However, MAPE is an unbiased statistic where a smaller value indicates a better predictability of the model (Galavi et al, 2013). Galavi et al (2013) built ARIMA and Adaptive Neuro-Fuzzy Inference System (ANFIS) models to forecast one-day-ahead water level in the Klang River. Their final ARIMA and ANFIS models had MAPE values of 0.337 and 0.404, respectively. All versions of our model have a lower MAPE than Galavi et al, suggesting that our parameter selection and hyperparameter tuning produced relatively effective ML models.

CONCLUSION

The ARIMA and LSTM models forecast general 30-day trends effectively. All versions of our models struggle to capture extreme discharge fluctuations, which is likely a cause of model scoring during our validation steps. The groundwater-absent LSTM could be used to forecast general 30-day discharge for ecological modeling purposes, but these models should not be used for flood warning. Most of our results suggest the groundwater did not improve model forecast quality. However, there are alternative methods for including groundwater in discharge forecasting. For example, groundwater could also be forecasted using an ARIMA model, and the results of that forecast can be an input into the primary discharge forecast model.

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APPENDIX

Table 2: Monitor identification codes and coordinates. Monitor types are abbreviated: S = stream gauge, G = groundwater monitor, and W = weather station.

Site	S	S coordinates	G	G coordinates	W	W coordinates
A	USGS-01571184	40.0309245327505, -77.3085949472522	USGS-400209077183301	40.0352777777778, -77.3080833333333	USC00360656	39.93468, -77.25459
B	USGS-01567000	40.4784167735445, -77.1291483971583	USGS-402735077100901	40.4604444444444, -77.1686944444445	USC00364992	40.5864, -77.5692
C	USGS-01550000	41.418058820116904, -77.0329971470846	USGS-420710077052101	41.4079166666667, -76.99475	USW00014778	41.24295, -76.92173

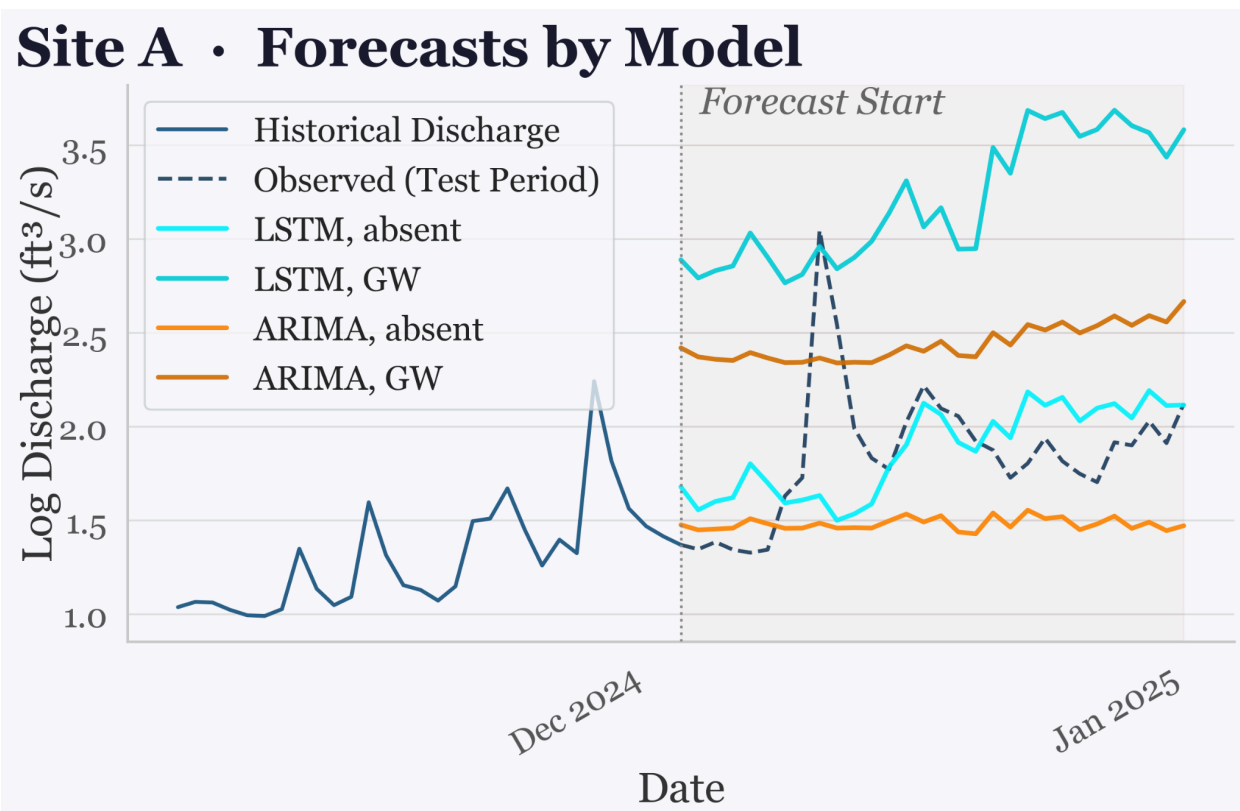

Figure 2: Forecast prediction lines for stream discharge at Site A. The LSTM absent model performed the best out of any model across all of the sites (RMSE = 0.398).

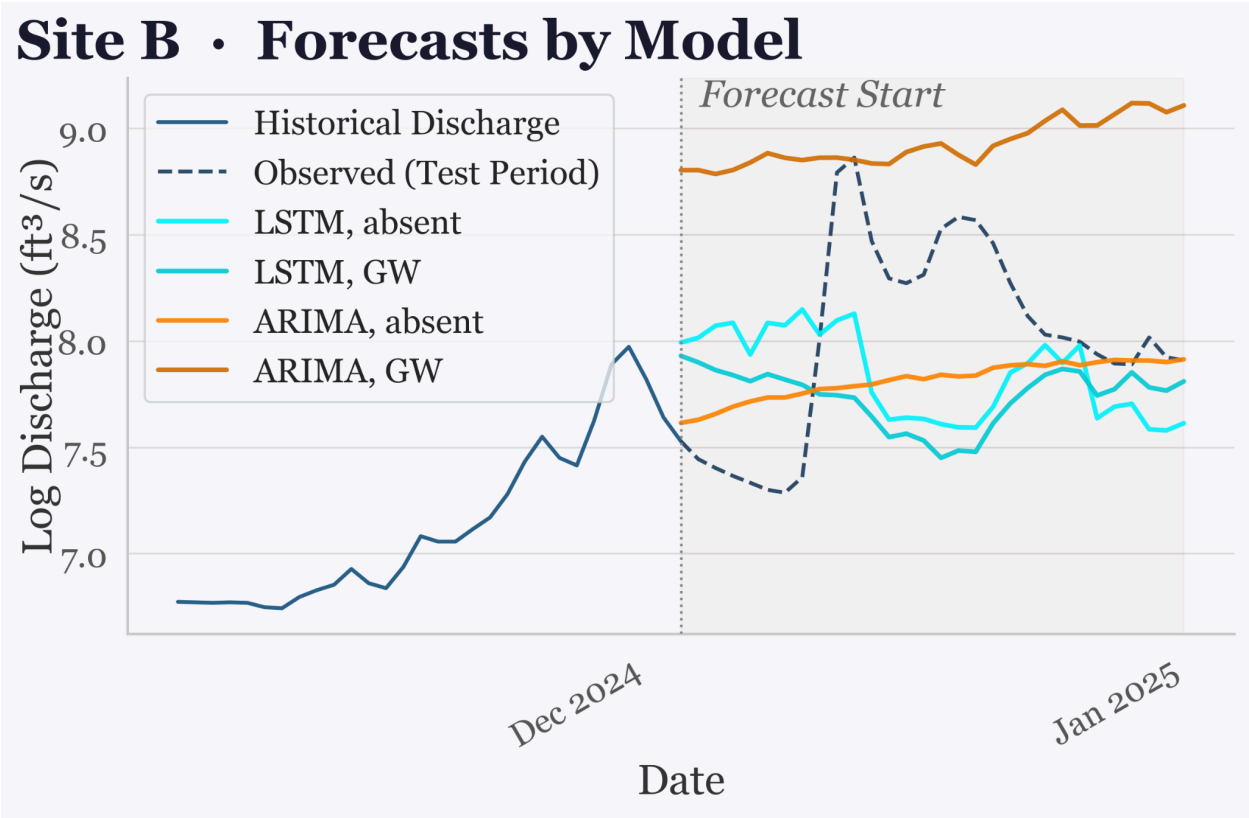

Figure 3: Forecast prediction lines for stream discharge at Site B. ARIMA absent (RMSE = 0.463) and LSTM absent (RMSE = 0.494) performed well. Both models underforecasted during high discharge events.

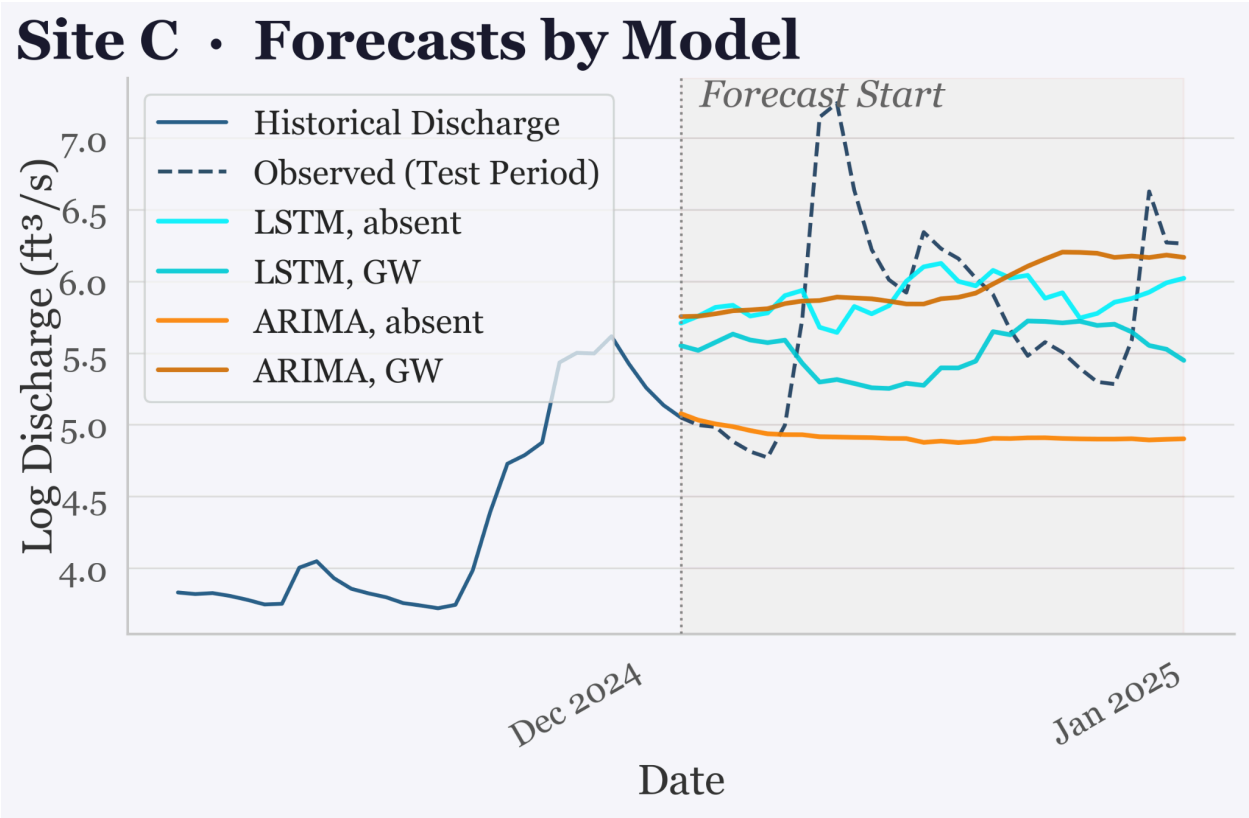

Figure 4: Forecast prediction lines for stream discharge at Site C. LSTM absent (RMSE = 0.664) and ARIMA groundwater (RMSE = 0.687) performed well, but neither captured the extreme discharge event.

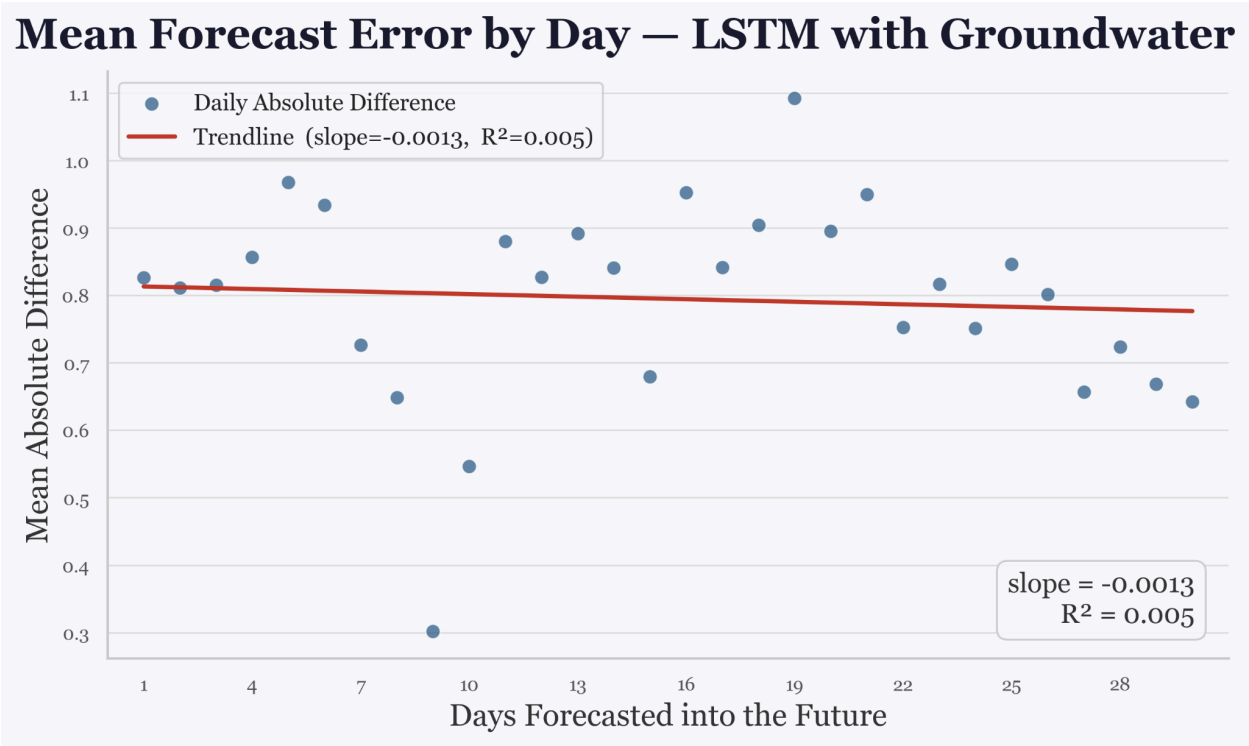

Figure 5: Forecast error averaged across all sites for the given number of days into the future. The negative slope (-0.0013) indicates forecast strength did not decay. By the nonlinear shape of the graph, forecast accuracy does not appear to be a function of days into the future but a result of poorly predicted extreme discharge events.

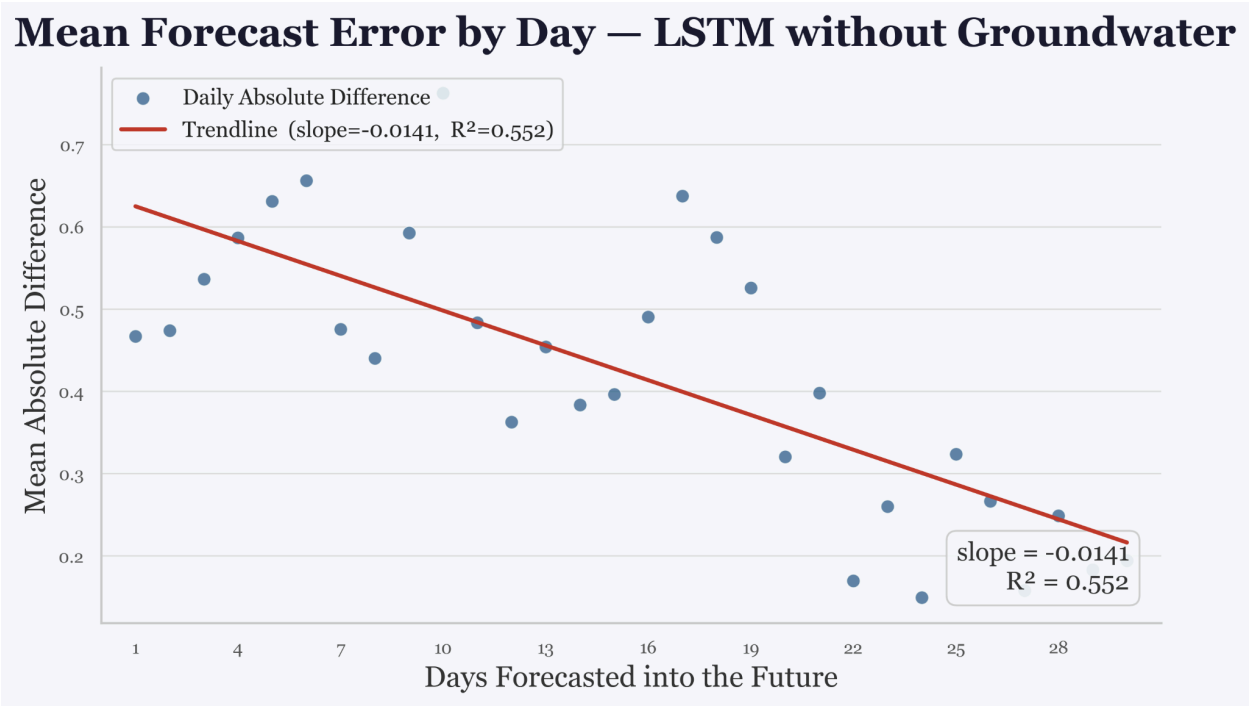

Figure 6: Forecast error averaged across all sites for the given number of days into the future. The negative slope (-0.0141) indicates forecast strength did not decay. The R^2 value is relatively strong, which also indicates the relatively consistent accuracy of the LSTM absent model.

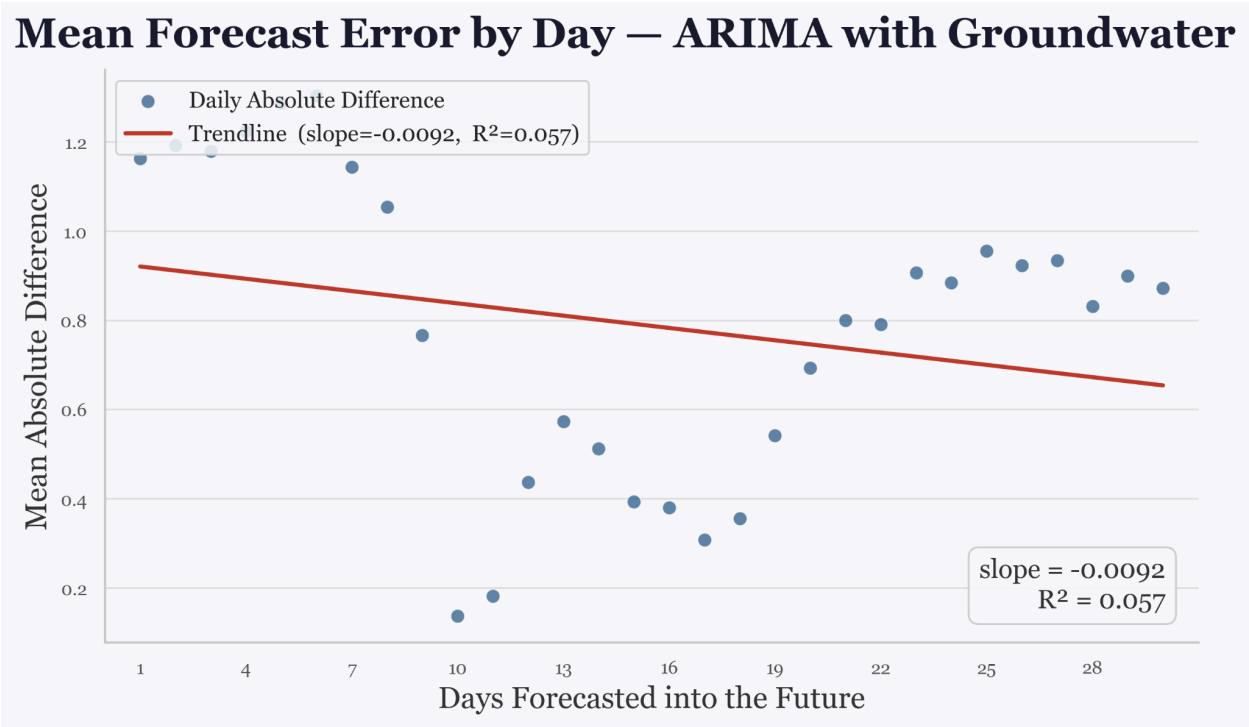

Figure 7: Forecast error averaged across all sites for the given number of days into the future. The negative slope (-0.0092) indicates forecast strength did not decay. By the nonlinear shape of the graph, forecast accuracy does not appear to be a function of days into the future but a result of extreme discharge events.

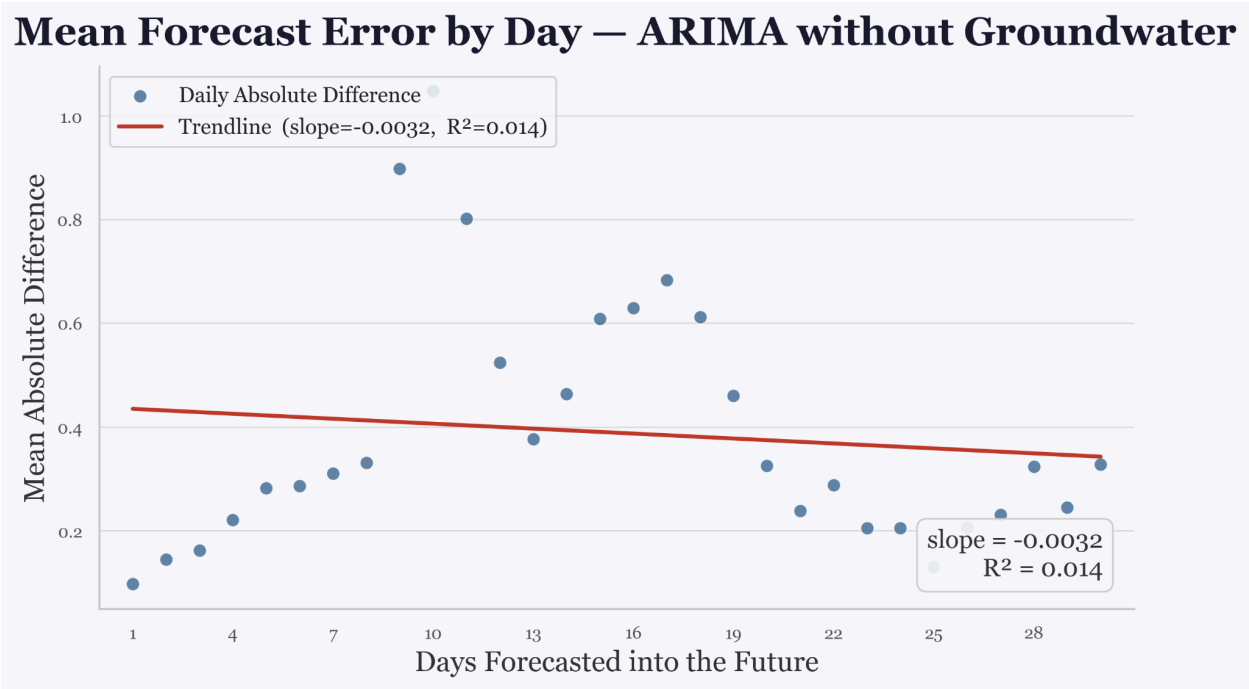

Figure 8: Forecast error averaged across all sites for the given number of days into the future. The negative slope (-0.0032) indicates forecast strength did not decay. By the nonlinear shape of the graph, forecast accuracy does not appear to be a function of days into the future but a result of extreme discharge events.